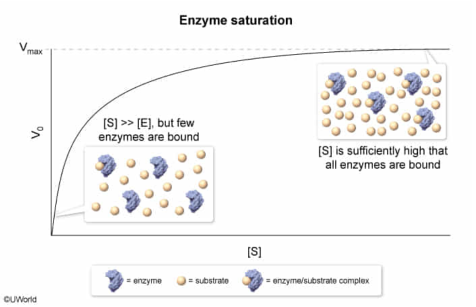
Which statement accurately describes a substrate in the presence of an enzyme operating at V_{max}? Only when the enzyme operates at V_{max} is the substrate:

- ✔ (selected) A. bound to every enzyme active site in solution. [73%]
- B. present at a significantly higher concentration than the enzyme. [19%]
- C. stabilized in the transition state. [3%]
- D. converted to product by a first-order reaction. [3%]

Correct
72% answered correctly

Explanation:



The **maximum possible reaction rate V_{max}** of an enzymatic reaction is **limited by the number of enzyme active sites** present in solution. A Michaelis-Menten graph depicts the rate of an enzymatic reaction V_0 as a function of substrate concentration [S] when the concentration of enzyme [E] is fixed. As [S] increases, the number of active sites bound by substrate also increases, leading to a faster reaction rate.

When [S] is sufficiently high, all enzyme active sites are bound (saturating conditions) and V_{max} is achieved. In contrast, when [S] is not saturating, some enzyme active sites are not bound by substrate and the reaction proceeds at a rate that is slower than V_{max}. Accordingly, only when the enzyme is operating **at V_{max}** does **substrate occupy every active site** in solution.

(Choice B) In a Michaelis-Menten graph, [S] is significantly higher than [E] at *every* substrate concentration measured, not just under saturating conditions. Having [S] that is significantly higher than [E] does *not* necessarily mean that all active sites are occupied, so the fact that [S] is significantly higher than [E] does *not* necessarily mean that V_{max} is achieved.

(Choice C) Enzymes catalyze reactions by stabilizing the transition state between substrate and product. This occurs *regardless* of whether or not sufficient substrate is present to achieve V_{max}.

(Choice D) In first-order reactions, the reaction rate increases linearly with substrate concentration. In Michaelis-Menten kinetics, reactions are approximately first-order at *low* substrate concentrations. However, under the saturating conditions required to achieve V_{max}, the reaction follows approximately zero-order kinetics, in which the rate does not increase with increasing substrate concentration.

Educational objective:
V_{max} is the maximum possible rate of an enzymatic reaction when a given concentration of enzyme is present. In a Michaelis-Menten graph, [S] is significantly larger than [E] at every [S] measured. However, V_{max} occurs only when the concentration of substrate [S] is high enough that all enzyme active sites are bound.

Time spent: 0 sec
Subject: Biochemistry

QID: 402438
System: Enzymes